

Education

- 2019–2025 **Doctoral degree (Dr. rer. nat.) in Parallel Programming**, *University of Kassel, Germany*
Dissertation: Load Balancing and Fault Tolerance for Asynchronous Many-Task Programs
- 2018–2019 **Master in Computer Science**, *University of Kassel, Germany*
Thesis: Design and Evaluation of a Work Stealing-Based Fault Tolerance Scheme for Task Pools
- 2014–2018 **Bachelor in Computer Science**, *University of Kassel, Germany*
Thesis: An Asynchronous Backup Scheme Tracking Work-Stealing for Reduction-Based Task Pools
- 2011–2013 **High School**, *Reichspräsident-Friedrich-Ebert-Schule, Fritzlar, Germany*
Specialization: Information Technology

Experience

- 2025–today **Postdoctoral Researcher**, *University of Kassel, Germany, Research Group Programming Languages/Methodology*
Research and teaching as a postdoctoral researcher
- 2019–2025 **Research Assistant**, *University of Kassel, Germany, Research Group Programming Languages/Methodology*
Research and teaching as a PhD candidate
- 2019 **Student Assistant**, *University of Kassel, Germany, Research Group Programming Languages/Methodology*
Assisting with scientific publications
- 2018 **Part time in IT**, *Hospital zum Hl. Geist gGmbH, Fritzlar, Germany*
Administrating Linux servers; Maintaining and updating IGEL thin clients; Extending a photo transfer software I wrote for the hospital in the past; Helpdesk support on first- and second level; Installing a new intranet web platform
- 2018 **Internship in IT**, *Hospital zum Hl. Geist gGmbH, Fritzlar, Germany*
Three months as requirement for my bachelor's degree
- 2015–2018 **Student Assistant**, *University of Kassel, Germany, Research Group Programming Languages/Methodology*
Grading of student homework, Assisting with the DFG research project "Fehlertoleranz und Ressourcen-Elastizität für globale Taskpools" and assisting publications beginning from 2017
- 2015 **Self-employed**, *Development and running a web platform for advertising vacation rentals*

2011–2012 **Internship in IT**, Hospital zum Hl. Geist gGmbH, Fritzlar, Germany
One year as part of highschool; Development of a photo transfer software for IGEL thin clients, and plugins to control HP switches from the helpdesk web platform GLPI

Publications and Talks

- 2026 **M. Reitz**, *Asynchronous Many-Task Programming for Space Edge Computing*; Presentation at the Workshop on Parallel Algorithms, Computer Architectures, and System Software (PARS), German Informatics Society (GI)
- 2026 **R. Nather, C. Fohry, and M. Reitz**, *Protecting Futures against Silent Data Corruption - Efficient Task Replication for Dynamic Data Dependencies*; International Conference on Parallel Processing and Applied Mathematics (PPAM), Workshop on Asynchronous Many-Task Systems for Exascale (AMTE), to appear
- 2026 **M. Reitz, D. Chenet, and J. Posner**, *Work Stealing for the 2D-Mesh Topology of Satellite Constellations in Low Earth Orbit*; International Conference on Parallel Processing and Applied Mathematics (PPAM), Workshop on Asynchronous Many-Task Systems for Exascale (AMTE), to appear
- 2026 **M. Reitz**, *Promise-Future Synchronization for Cluster Asynchronous Many-Task Runtimes via MPI One-Sided Communication*; Euro-Par 2026, poster paper, to appear
- 2026 **M. Reitz and C. Fohry**, *Silent Data Corruption Protection through Efficient Task Replication*; Proceedings of the Workshop on Asynchronous Many-Task Systems and Applications (WAMTA)
- 2026 **T. R. Lahnor, M. Reitz, J. Posner, and P. Diehl**, *Exploring Performance-Productivity Trade-offs in AMT Runtimes: A Task Bench Study of Itoyori, ItoyoriFBC, HPX, and MPI*; Proceedings of the Workshop on Asynchronous Many-Task Systems and Applications (WAMTA)
- 2025 **M. Reitz**, *Load Balancing and Fault Tolerance for Asynchronous Many-Task Programs*; University of Kassel, Germany, **Dissertation**
- 2025 **M. Reitz, J. Hundhausen, and C. Fohry**, *Fail-stop Failure Protection for Coordinated Work Stealing of Tasks that Communicate through Futures*; Workshop on Asynchronous Many-Task Systems and Applications (WAMTA), pp. 44–55
- 2025 **M. Reitz and J. Posner**, *Stackless vs. Stackful Coroutines: A Comparative Study for RDMA-based Asynchronous Many-Task (AMT) Runtimes*; Supercomputing, Parallel Applications Workshop (PAW-ATM), pp. 1341–1350
- 2024 **R. Nather, M. Reitz, and C. Fohry**, *Distributed, Resilient and In-Memory Storage of Key-Value Data for HPC*; WIP talk, Supercomputing: International Parallel Data System Workshop (PDSW)

- 2024 **M. Reitz, B. Gerhards, J. Hundhausen, and C. Fohry**, *Investigating the Performance Difference of Task Communication via Futures or Side Effects; International European Conference on Parallel and Distributed Computing (EuroPar), Workshop on Asynchronous Many-Task Systems for Exascale (AMTE)*, pp. 255–267
- 2024 **M. Reitz and C. Fohry**, *Task-Level Checkpointing and Localized Recovery to Tolerate Permanent Node Failures for Nested Fork-Join Programs in Clusters; SN Computer Science* 5, 320
- 2024 **M. Reitz**, *Fault Tolerance and Locality-Aware Work Stealing for Dynamically Generated Dependent Tasks on Clusters; HKHLR report*
- 2023 **M. Reitz and C. Fohry**, *Task-Level Checkpointing for Nested Fork-Join Programs using Work Stealing; International European Conference on Parallel and Distributed Computing (EuroPar), Workshop on Asynchronous Many-Task Systems for Exascale (AMTE)*
- 2023 **M. Reitz, K. Hardenbicker, T. Werner, and C. Fohry**, *Lifeline-based Load Balancing Schemes for Asynchronous Many-Task Runtimes in Clusters; Parallel Computing (PARCO) (Special Issue), Vol. 116*
- 2022 **M. Reitz, K. Hardenbicker, and C. Fohry**, *Comparison of Load Balancing Schemes for Asynchronous Many-Task Runtimes; International Conference On Parallel Processing and Applied Mathematics (PPAM), Workshop on Language Based Parallel Programming (WLPP)*
- 2022 **J. Posner, M. Reitz, and C. Fohry**, *Task-Level Resilience: Checkpointing vs. Supervision; International Journal of Networking and Computing (IJNC) (Special Issue)*, pp. 47–72
- 2021 **M. Reitz**, *Load Balancing Policies for Nested Fork-Join; Poster. Extended abstract in Proceedings IEEE International Conference on Cluster Computing (CLUSTER)*, pp. 817–818
- 2021 **M. Reitz**, *Task-Level Checkpointing for Nested Fork-Join Programs; Poster. PhD Forum International Parallel and Distributed Processing Symposium (IPDPS)*
- 2021 **J. Posner, M. Reitz, and C. Fohry**, *Checkpointing vs. Supervision Resilience Approaches for Dynamic Independent Tasks; IEEE International Parallel and Distributed Processing Symposium, Workshop on Advances in Parallel and Distributed Computational Models (APDCM)*, pp. 556–565
- 2020 **J. Posner, M. Reitz, and C. Fohry**, *A Comparison of Application-Level Fault Tolerance Schemes for Task Pools; Future Generation Computer Systems (FGCS) (Special Issue)*, pp. 119–134
- 2018 **J. Posner, M. Reitz, and C. Fohry**, *Comparison of the HPC and Big Data Java Libraries Spark, PCJ and APGAS; Supercomputing, Parallel Applications Workshop (PAW-ATM)*, pp. 11–22
- 2018 **C. Fohry, J. Posner, and M. Reitz**, *A Selective and Incremental Backup Scheme for Task Pools; Int. Conf. on High Performance Computing & Simulation (HPCS)*, pp. 621–628

Teaching

Lectures and Exercise Sessions

- 2020–2022, 2026–today **Algorithms and Data Structures**, *Preparing homework for students; leading exercise sessions; designing exams (“Testate”) that students must pass to qualify for the final exam*
- 2020–2023 **Introduction to Parallel Processing**, *Lecturing students about programming GPUs using CUDA (2020–2021) and OpenMP (2022–2023)*
- 2023 **Seminar: Current State and Trends in High Performance Computing**, *Leading the seminar*
- 2020–2022 **Introduction to Computer Science**, *Preparing homework for students and leading exercise sessions*
- 2021 **Seminar: Task-based Parallel Programming Systems**, *Leading the seminar*

Supervised Master Theses

- 2026 **Integration and Evaluation of a Work-Stealing Scheme in a Task-based Parallel Programming System with Futures**
- 2024 **Integration of the Asynchronous Many-Task Runtime Itoyori into the Benchmark Suite TaskBench**

Supervised Bachelor Theses

- 2026 **Comparative Analysis of Performance and Programmability of Task-based Parallel Programming Systems: ItoyoriFBC vs. Legion**
- 2026 **Design and Evaluation of a Load Balancing Scheme for Space Edge Computing**
- 2023 **Comparison of Randomized vs. Lifeline-based Cooperative Work Stealing with Nested Fork-Join Programs**
- 2023 **Comparison of Randomized vs. Lifeline-based Cooperative Work Stealing with Independent Tasks**
- 2022 **Coordinated Work Stealing in the Parallel Programming Environment GLB**
- 2022 **Parallel Evaluation of Neural Networks using Task-based Parallel Programming**
- 2022 **Integration of the Parallel Programming Environment GLB into the Benchmark Suite TaskBench**
- 2022 **Locality-optimized Work-Stealing for Task-based Parallel Programming Environments in Clusters**
- 2021 **Logging and Visualization of Runtime Behaviour of Distributed Taskpools**

Supervised Master Projects

- 2023 **Design and Implementation of a Preprocessor for an experimental Parallel Programming System**
- 2023 **Comparison between Randomized vs. Lifeline-based Coordinated Work-Stealing**

Supervised Bachelor Projects

- 2026 **Porting and Benchmarking of ItoyoriFBC using the NAS Benchmark Suite**
- 2026 **Locality-awareness for Work Stealing of Tasks with Futures**
- 2023 **Implementation of Parallel Backtracking in Distributed Memory in the Programming Language Julia**
- 2022 **Implementation and Comparison of Current SplitQueue Variants with Java Concurrent Queues**
- 2022 **Implementation of Nbody-Simulation as a Benchmark for a Task-based Parallel Programming Environment**
- 2022 **Assessment of the RDMA-based Serialization-free Networking library Naos**

Service to Profession

- 2026 **International Conference on Parallel Processing and Applied Mathematics (PPAM), Workshop on Asynchronous Many-Task Systems for Exascale (AMTE), Program Committee Member and Reviewer**
- 2026 **Workshop on Asynchronous Many-Task Systems and Applications (WAMTA), Invited Reviewer**
- 2024–2026 **Supercomputing Conference, Invited Reviewer for the Student Volunteer Program and HPC Immersion Program**
- 2024 **Journal of Parallel and Distributed Computing (JPDC), Invited Reviewer (two reviews)**
- 2024 **Supercomputing Conference (SC24), Lead Student Volunteer, Atlanta, USA**
- 2019–2023 **Supercomputing Conference, Student Volunteer, Denver, USA; Virtual; St. Louis, USA (Hybrid-Virtual); Dallas, USA; Denver, USA**

Innovation and Transfer

- 2026 **UNIKAT Ideas Competition, Audience Award and third place (1,500 Euro prize money), University of Kassel**
For “Second Nose”, a smartphone dongle and app supporting people with reduced or absent sense of smell

External funding

- 2020–2026 **Goethe Cluster Frankfurt, accepted project proposal granting (annually) 300,000 to 800,000 core hours, title: Fehlertoleranz und Ressourcen-Elastizität für globale Taskpools**
- 2023–2024 **Lichtenberg Cluster Darmstadt, accepted project proposal granting 360,000 core hours, title: Fault Tolerance and Locality-Aware Work Stealing for Dynamically Generated Dependent Tasks on Clusters**

Languages

German	Native
English	Fluent
Japanese	Good (JLPT N2 certified)
French	Beginner

Mia Reitz



Kassel, Germany October 7, 2026